Week 12-13 Office Hour Notes

12.0 Errata

%mention the most recent correction on 1/7/25 involving incorrect mention of axiom group 6 as $x = y \to \left(\phi \to \phi_y^x\right)$

12.1 Logical Axioms

Now we move on to the "proof system" of first order logic, where the symbol we use is $\Gamma \vdash \phi$ instead of $\Gamma \vDash \phi$. The latter notion refers to truth implication for all structures paired with variable assignment functions, and the former notion is going to refer to the ability to "prove" ϕ from Γ. To do this, we formalize the proof system. We start formalizing the proof system by naming what we call the "logical axioms".

In the beginning of *Section 2.4 of Enderton* and this week's powerpoint, we discuss what we call logical axioms, which we define as follows:

Definition 12.1.1.

1. Given a wff ψ, we state that φ is a **generalization** of ψ if for $n \geq 0$ and arbitrary variables $x_1, \ldots, x_n$, we have

$\varphi := \forall x_1 \cdots \forall x_n \psi.$

2. We consider the set Λ of wff's the call the set of **logical axioms**, consisting of *generalizations* of wff's of the following form (note that we define this all in the subsequent definitions) given that α, β are wff's

(i) Tautologies (corresponds to sentential logic, as we'll see in a moment)
(ii) Formulas of the form $\forall x \alpha \to \alpha_t^x$, where t is subsitutable for x in α.
(iii) $\forall x(\alpha \to \beta) \to \left(\forall x \alpha \to \forall x \beta\right)$
(iv) $\alpha \to \forall x \alpha$ if x does not occur free in α.

If the language indludes equality, then we include

(v) $x = x$
(vi) $x = y \to \left(\phi \to \phi'\right),$

where ϕ is atomic and ϕ' is obtained by replacing x in zero or more (but not necessarily all) places by y.

Note that *(iii)*, *(iv)*, *(v)*, and *(vi)* are straightforward constructions, but *(i)* and *(ii)* needs further elaboration, which we shall now do:

Definition 12.1.2.

1. A **prime formula** is defined to be either an atomic formula (of the form $= t_1 t_2$ or $Pt_1 \ldots t_n$ for a n-ary predicate symbol P and terms $t_1, \ldots, t_n$ in the language) or those of the form $\forall x \alpha$. Note that all well-formed formulas are made up of prime formulas and we have the following structural induction property on all wff formulas, based on prime formulas

(a) (base case) ϕ is a prime formula.
(b) (inductive step) If ϕ is of the form $\neg\psi$ or $\psi \to \varphi$.

2. We call a first order wff ϕ a **tautology** if we take the sentential form Sen(ϕ) of ϕ and find that Sen(ϕ) is a tautology, i.e., we have $\vDash$ Sen(ϕ) in sentential logic, defined by replacing each prime formulas with a distinct sentence symbol.

More rigorously, we define Sen(ϕ) by enumerating every prime formula $\{\phi_n\}_{n \in \mathbb{N}}$ and then proceed by recursion (recall the use of **Theorem 4.2.3** on the Week 4 Office Hours) on prime formulas as follows:

(a) If ϕ is itself prime, i.e. ϕ is ϕ_n for some $n \in \mathbb{N}$, then Sen(ϕ) is the sentence symbol A_n.
(b) If ϕ is $\neg\psi$, then Sen($\neg\psi$) = $\neg$Sen(ψ).
(c) If ϕ is $\psi \to \varphi$, then Sen(ϕ) is Sen(ψ) $\to$ Sen(φ).

and if Sen(ϕ) is a tautology, i.e. $\vDash$ Sen(ϕ), then ϕ by **Definition 12.1.1** is logical axiom and we have $\phi \in \Lambda$

Example 12.1.3. If ϕ is $\forall x \forall y(x = y) \vee \neg\forall x \forall y(x = y)$, then note first of all that this is shorthand in the language for

$$\neg\forall x \forall y(x = y) \to \neg\forall x \forall y(x = y)$$

To define Sen(ϕ), note that the prime formula is $\forall x \forall y(x = y)$, so we subsitute $\forall x \forall y(x = y)$ for A_1 and we get

$\text{Sen}(\phi) = \neg A_1 \to \neg A_1$

Since

$\neg A_1 \to \neg A_1 \vDash\Dashv \neg\neg A_1 \vee \neg A_1 \vDash\Dashv A_1 \vee \neg A_1,$

which is just the *Law of Excluded Middle*, we find $\vDash \text{Sen}(\phi)$, hence ϕ is a tautology and we have $\phi \in \Lambda$.

Next we want to talk about what it means for a term t to be substitutable for x in a wff α, but first we have to define α_x^t, which we shall do by structural recursion on wff's.

If α is an atomic formula of the form $= t_1 t_2$ or $Rt_1 \dots t_n$ for any $t_1, \dots, t_n \in \text{Term}_{\mathcal{L}}$ and n-ary predicate symbol $R \in \mathcal{R}$, then

$\alpha_t^x := \ = (t_1)_t^x (t_2)_t^x$ or $R(t_1)_t^x \dots (t_n)_t^x$

If α is of the form:

$\neg\alpha_1$, then $\alpha_t^x := \neg(\alpha_1)_t^x$
$\alpha_1 \to \alpha_2$, then $\alpha_t^x := (\alpha_1)_t^x \to (\alpha_2)_t^x$
$\forall y \alpha_1$, then $\alpha_t^x := \begin{cases} \forall y \alpha_1 & \text{if } x = y \\ \forall y\left((\alpha_1)_t^x\right) & \text{if } x \neq y \end{cases}$

Recall that in *Exercise 2.4.1 of Enderton* (which has been discussed in both the Week 10 and 11 dicussion forums) we defined $(u)_t^x$ for any term $u \in \text{Term}_{\mathcal{L}}$ by structural recursion on u.

Definition 12.1.4. We define a term t to be **substitutable** for x in α if it satisfies any of the following (inductive conditions):

1. For atomic α, t is (always) substitutable for x in α.
2. If α is $\neg\alpha_1$ and t is subsitutable for x in α_1.
3. If α is $\alpha_1 \to \alpha_2$ and t is subsitutable for x in both α_1.and α_2
4. If α is $\forall y \alpha_1$ and either
(a) x does not occur free in α_1
(*b)* y does not occur in t and t is subsitutable for x in α

So to conclude the intuitive idea is that t is substituable for x in α preciesely when no variable is involved in t that is quantified over α.

12.2 Nonogical Axioms, Modens Ponens, and Hilbert Calculus

So to talk about a proof system, we next deal with both the nonlogical axioms, which consists of a set of wff's Γ, what we call inference rules, i.e., what we are able to logically infer from the formulas that either we've already infered or are logical/nonlogical axioms.

In the proof system aka **Hilbert Calculus** that we plan to examine in this course, we consider only the inference rule that we call *Modens Ponens* (some texts call it "Arrow Elimination" $\to E$) where if we deduce both α and $\alpha \to \beta$, we can infer β.

$\alpha \quad \alpha \to \beta$
$------- \to E$
β

Definition 12.2.1. We consider a set Γ to be **closed under Modens Ponens** if for every wff α, β

$\alpha, \alpha \to \beta \in \Gamma \Longrightarrow \beta \in \Gamma.$

Next, we define what a deduction of ϕ from Γ is:

Definition 12.2.2. We call a sequence $\langle \phi_1, \ \dots, \phi_n \rangle$ a **(Hilbert) deduction** of $\phi := \phi_n$ from Γ if for all ϕ_k $(1 \leq k \leq n)$, either

1. $\phi_k \in \Lambda \cup \Gamma$.
2. ϕ_k can be concluded from previous entries in the deduction from *Mondens Ponens*, i.e., if ϕ_j is $\phi_i \to \phi_k$, for $i, j < k$.

we state that $\Gamma \vdash \phi$ if there exists a deduction $\langle \phi_1, \ \dots, \phi_n \rangle$ of $\phi := \phi_n$ from Γ.

Example 12.2.3.

(i) Let's say $\mathcal{L} := \{P\}$ for some unary predicate symbol P, and we have $\Gamma := \left\{ Pv_1, Pv_1 \to Pv_2 \right\}$. We find that $\Gamma \vdash Pv_2$ since we find $\langle Pv_1, Pv_1 \to Pv_2, Pv_2 \rangle$ is a deduction of Pv_2 from Γ, since we find that $Pv_1, Pv_1 \to Pv_2 \in \Gamma$ and we can deduce Pv_2 from the previous entries $Pv_1, Pv_1 \to Pv_2$ from Mondens Ponens

$$\begin{array}{l} Pv_1 \;\; Pv_1 \to Pv_2 \\ \text{-------------} \to E. \\ \quad Pv_2 \end{array}$$

(ii)
%GIVE MORE EXAMPLE OF DEDUCTION

Next time, we'll prove the following theorem about natural deduction, defining

Th(Γ) := $\left\{ \text{wff's } \phi : \, \Gamma \vdash \phi \right\}$.

Theorem 12.2.3. Th(Γ) is the smallest set containing $\Gamma \cup \Lambda$ closed under Modens Ponens. Moreover, the following are equivalent:

1. Γ contains Λ and is closed under Modens Ponens.
2. Γ is closed under deduction, i.e., if ϕ is a wff such that a deduction $\langle \phi_1, \, \ldots, \phi_n \rangle$ of $\phi := \phi_n$ exists from Γ, then $\phi \in \Gamma$.
3. There exists a set Γ_0 such that Th(Γ_0) $= \Gamma$.

Proof.

Let's prove $1. \iff 2. \iff 3.$ using $1. \implies 2. \implies 3.$ First, note that $2. \implies 3.$ is immediate from the fact that Th(Γ) $= \Gamma$ from the fact that $\Gamma \subset$ Th(Γ) since $\langle \gamma \rangle$ is a deduction of every $\gamma \in \Gamma$ and conversely we have $\Gamma \supset$ Th(Γ) from the fact that for every $\phi \in$ Th(Γ) has a deduction $\langle \phi_1, \, \ldots, \phi_n \rangle$ from Γ to $\phi_n := \phi$, implying by hypothesis that $\phi \in \Gamma$. Next, note that $3. \implies 1.$ is immediate by choosing Γ_0 such that Th(Γ_0) $= \Gamma$ and noting by hypothesis that

$\Lambda \subset$ Th(Γ_0) $= \Gamma$,

and that we have closure under Modens Ponens since for every $\alpha, \alpha \to \beta \in \Gamma =$ Th(Γ_0) implies there exists a deduction $\langle \alpha_1, \, \ldots, \alpha_n \rangle$ from Γ_0 to $\alpha_n := \alpha$ and $\langle {\alpha_1}', \, \ldots, {\alpha_m}' \rangle$ from Γ_0 to ${\alpha_m}' := \alpha \to \beta$. It then follows that $\langle \alpha_1, \, \ldots, \alpha_n, {\alpha_1}', \, \ldots, {\alpha_m}', \beta \rangle$ is a deduction from Γ_0 to β, where we can conclude all of $\alpha_1, \, \ldots, \alpha_n, {\alpha_1}', \, \ldots, {\alpha_m}'$ from the deductions $\langle \alpha_1, \, \ldots, \alpha_n \rangle, \langle {\alpha_1}', \, \ldots, {\alpha_m}' \rangle$ being defined and conclude β from Modens Ponens since we have $\alpha_n := \alpha$ and ${\alpha_m}' := \alpha \to \beta$. We then conclude that $\beta \in$ Th(Γ_0) $= \Gamma$ since $\Gamma_0 \vdash \beta \implies \beta \in$ Th(Γ_0).

it remains to show $1. \implies 2.$, which follows from taking a wff ϕ such that $\Gamma \vdash \phi$, choosing a deduction $\langle \phi_1, \ldots, \phi_n \rangle$ from Γ to $\phi_n := \phi$ and showing that $\phi \in \Gamma$ by induction on the length $n \geq 1$ of the deduction $\langle \phi_1, \ldots, \phi_n \rangle$ and use of closure of Modens Ponens. Details I'll leave to the reader.

To show that Th(Γ) is the smallest set containing $\Gamma \cup \Lambda$ and is closed under Modens Ponens, we first note by conditions 1. and 3. that Th(Γ) contains containing $\Gamma \cup \Lambda$ and is closed under Modens Ponens. Then take the set $\mathbb{C}$ of all sets of wff's Σ closed under Modens Ponens and show that

$$\bigcap_{\Sigma \in \mathbb{C}} \Sigma = \text{Th}(\Gamma).$$

Details I'll once again leave to the reader. □

%TALK ABOUT HOW THE SET OF THEOREMS IS AN INDUCTIVE SET, BUT ISN'T "FREELY GENERATED"

13.0 Errata

13.1 Deriving Proof/Inference Strategies

%STATE THE INFERENCE RULES FROM THE INFERENCE RULES LIST ON WIKIPEDIA LINKED HERE

Theorem 13.1.1. *(Rule T aka Lemma 24C of the book)* If $\Gamma \vdash \alpha_1, \ldots, \Gamma \vdash \alpha_n$ and $\left\{\text{Sen}(\alpha_1), \ldots, \text{Sen}(\alpha_n)\right\}$ tautologically implies Sen(β) (i.e. we have $\left\{\text{Sen}(\alpha_1), \ldots, \text{Sen}(\alpha_n)\right\} \vDash \text{Sen}(\beta)$ in sentential logic) then $\Gamma \vdash \beta$.

Theorem 13.1.2. *(Deduction Theorem for Deduction)* $\Gamma; \gamma \vdash \varphi \iff \Gamma \vdash \gamma \rightarrow \varphi$. (Note that is $\rightarrow I$)

Proof.
%LOOK UP PROOF IN BOOK AND COMPARE IT TO PROOF OFF THE DOME □

Note that the deduction theorem tells us that to prove $\Gamma \vdash \gamma \rightarrow \varphi$, it suffices (as it does when proving a corresponding statement in the metalanguage) to additionally add γ to our list of nonlogical axioms Γ (as what we call a hypothesis) and show that a proof exists between

$\Gamma; \gamma$ and φ, i.e. $\Gamma; \gamma \vdash \varphi$.

Additionally, we find from the *Deduction Theorem* that we can do the proof strategies such as the contrapositive and proof by contradiction that we can do in the metalanguage, as we state formally in the following corollaries:

Corollary 13.1.3. *(Contraposition; restatement of Corollary 24D of Enderton)*
$\Gamma; \varphi \vdash \psi \Longleftrightarrow \Gamma; \neg\psi \vdash \neg\varphi$.

Proof.

$\Longrightarrow$ Assume that $\Gamma; \varphi \vdash \psi$. Note first off that for any wff φ, ψ, we find that $(\varphi \to \psi) \to (\neg\psi \to \neg\varphi)$ (note this can be verified using truth tables) is a tautology. Next, we find by the Deduction Theorem and Modens Ponens that

$$\begin{aligned}
\Gamma; \varphi \vdash \psi &\Longrightarrow \Gamma \vdash \varphi \to \psi, \Gamma \vdash (\varphi \to \psi) \to (\neg\psi \to \neg\varphi) \\
&\Longrightarrow \Gamma \vdash \neg\psi \to \neg\varphi \\
&\Longrightarrow \Gamma; \neg\psi \vdash \neg\varphi.
\end{aligned}$$

$\Longleftarrow$ Similar to $\Longrightarrow$, except using the **Deduction Theorem** and **Modens Ponens** involving the tautology $(\neg\psi \to \neg\varphi) \to (\varphi \to \psi)$. ◻

Definition 13.1.4. We say that Γ is **consistent** if $\Gamma \nvdash \neg\beta \wedge \beta$, for some wff β.

Corollary 13.1.4. *(Reductio ad absurdum aka Proof by contradiction; restatement of Corollary 24E of Enderton)*

1. $(\neg I)$ If $\Gamma; \varphi$ is inconsistent, then $\Gamma \vdash \neg\varphi$.
2. $(\neg E)$ If $\Gamma; \neg\varphi$ is inconsistent, then $\Gamma \vdash \varphi$.

Proof. Left to the reader. Some hints will be given maybe in future draft. ◻

Proposition 13.1.5. The following are equivalent.

1. Γ is inconsistent.

2. $\Gamma \vdash \varphi$, for every wff φ

3. $\Gamma \vdash \neg\varphi \wedge \varphi$, for every wff φ.

4. $\Gamma \vdash \perp$, where $\perp := \forall v_1(\neg v_1 = v_1)$

Philisophical idea: We want a proof system that refutes certain things and in particular doesn't prove anything, since we want **Reducto Ad Absurdum** (proof by contradiction) to actually be meaningful in proving things, and same with any style of proof.

Example 13.1.6.

(i) Let's say $\Gamma := \{\neg\forall v_1 Pv_1, Pv_1, Pv_2, \ldots\}$. Then by **Corollary 13.2.5** (see the next section), we find Γ is consistent, since it's satisfiable because take $\mathfrak{A} := \left(|\mathfrak{A}|, P^{\mathfrak{A}}\right) |\mathfrak{A}| := \{0, 1\}, P^{\mathfrak{A}} := \{1\}$, define $s(v_n) := 1$, for all $v_n \in V$, Obviously $\vDash_{\mathfrak{A}} Pv_n[s]$, for all $n \geq 1$, however we also have $\vDash_{\mathfrak{A}} \neg\forall v_1 Pv_1[s]$, because we have $\nvDash_{\mathfrak{A}} \forall v_1 Pv_1$, because for $0 \in |\mathfrak{A}|$, we have $\nvDash_{\mathfrak{A}} Pv_1[s(v_1|0)]$, since $s(v_1|0)(v_1) = 0 \notin \{1\} = P^{\mathfrak{A}}$.

Equivalently, we have $\nvDash_{\mathfrak{A}} \forall v_1 Pv_1[s]$ iff there exists some $a \in |\mathfrak{A}|$ such that $\nvDash_{\mathfrak{A}} Pv_1[a]$, hence $\nvDash_{\mathfrak{A}} Pv_1[0]$, since $s(v_1|0)(v_1) = 0 \notin \{1\} = P^{\mathfrak{A}}$

(ii) Let $\Gamma := \{\forall v_1(\neg v_1 = v_1)\}$. We find that Γ is inconsistent, since the equality axiom gives us

$$\dfrac{\dfrac{\Gamma \vdash \forall x(x = x),\ \Gamma \vdash \forall x(x = x) \to y = y}{\Gamma \vdash y = y} \qquad \dfrac{\Gamma \vdash \forall x(\neg x = x),\ \Gamma \vdash \forall x(\neg x = x) \to \neg y = y}{\Gamma \vdash \neg y = y}}{\Gamma \vdash y = y \wedge \neg y = y,}$$

therefore Γ is inconsistent.

Additionally, we can show the following familiar inferences that we do in the metalanguage:

Corollary 13.1.5. We find that the following inferences can be established in the proof system:

1. $(\wedge I)$ if $\Gamma \vdash \varphi$ and $\Gamma \vdash \psi$, then $\Gamma \vdash \varphi \wedge \psi$.
2. $(\wedge E)$ If $\Gamma \vdash \varphi \wedge \psi$ then $\Gamma \vdash \varphi$ and $\Gamma \vdash \psi$.
3. $(\vee I)$ If $\Gamma \vdash \varphi$ or $\Gamma \vdash \psi$, then $\Gamma \vdash \varphi \vee \psi$.
4. $(\vee E)$ If $\Gamma; \varphi \vdash \alpha$ and $\Gamma; \psi \vdash \alpha$, then $\Gamma; \varphi \vee \psi \vdash \alpha$.
5. $(\forall I)$ If $\Gamma \vdash \varphi(x)$, where x is a variable that doesn't occur free in the hypotheses utilized in some deduction from Γ to $\varphi(x)$, then $\Gamma \vdash \forall x\varphi(x)$

6. $(\forall E)$ if $\Gamma \vdash \forall x\varphi$ and t is substitutable for x in φ, then $\Gamma \vdash \varphi^x_t$.
7. $(\exists I)$ If $\Gamma \vdash \varphi^x_t$ and t is substitutable for x in φ, then $\Gamma \vdash \exists x\varphi(x)$.
8. $(\exists E)$ If $\Gamma \vdash \exists x\varphi$, where x is a variable that doesn't occur free in the hypotheses utilized in some deduction from Γ to $\varphi(x)$, then $\Gamma \vdash \varphi(x)$,

Proof.

1. Note that $\varphi \wedge \psi$ is $\neg\left(\varphi \to \neg\psi\right)$, and we find using **Corollary 13.1.4 part 1** $(\neg I)$, we have

$$
\dfrac{\dfrac{\Gamma; \varphi \to \neg\psi \vdash \varphi \quad \Gamma; \varphi \to \neg\psi \vdash \varphi \to \neg\psi}{\Gamma; \varphi \to \neg\psi \vdash \neg\psi} \to E \qquad \Gamma; \varphi \to \neg\psi \vdash \psi}{\Gamma \vdash \varphi \wedge \psi} \neg I
$$

2. Left as an exercise to the reader.

3. Left as an exercise to the reader.

4. Note using truth tables that $\left(\varphi \to \alpha\right) \wedge \left(\psi \to \alpha\right) \to \left(\varphi \vee \psi \to \alpha\right)$ is a tautology,

φ	ψ	α	$\left(\varphi \to \alpha\right) \wedge \left(\psi \to \alpha\right)$	$\left(\varphi \vee \psi \to \alpha\right)$	$\left(\varphi \to \alpha\right) \wedge \left(\psi \to \alpha\right) \to \left(\varphi \vee \psi \to \alpha\right)$
F	F	F	T	T	T
F	F	T	F	F	T
F	T	F	F	F	T
F	T	T	T	T	T
T	F	F	F	F	T
T	F	T	T	T	T
T	T	F	F	F	T
T	T	T	T	T	T

so we have $\left(\varphi \to \alpha\right) \wedge \left(\psi \to \alpha\right) \to \left(\varphi \vee \psi \to \alpha\right) \in \Lambda$, hence

$$\Gamma \vdash \left(\varphi \to \alpha\right) \wedge \left(\psi \to \alpha\right) \to \left(\varphi \vee \psi \to \alpha\right).$$

Then using the Deduction Theorem $\left(\to I\right)$ and part 1 of this corollary $(\wedge I)$ we have

$$\Gamma; \varphi \vdash \alpha \qquad \Gamma; \psi \vdash \alpha$$

$$\text{-------} \to I \quad \text{-------} \to I$$

$$\Gamma \vdash \varphi \to \alpha \qquad \Gamma \vdash \psi \to \alpha$$

$$\text{----------------------} \wedge I$$

$$\Gamma \vdash \left(\varphi \to \alpha\right) \wedge \left(\psi \to \alpha\right) \qquad \Gamma \vdash \left(\varphi \to \alpha\right) \wedge \left(\psi \to \alpha\right) \to \left(\varphi \vee \psi \to \alpha\right)$$

$$\text{--} \to E$$

$$\Gamma \vdash \varphi \vee \psi \to \alpha,$$

and we conclude using the Deduction Theorem that $\Gamma; \varphi \vee \psi \vdash \alpha$.

5. Suppose $\Gamma \vdash \varphi(x)$, where x is a variable that doesn't occur free in the hypotheses utilized in some deduction D from Γ to $\varphi(x)$. Let $\{\varphi_1, \ldots, \varphi_n\} \subset \Gamma$ be the set of all wff's utilized in the deduction D that are determined from $\varphi_1, \ldots, \varphi_n \in \Gamma$, do not have x occuring free in each formula, and by hypothesis do not follow from any of other possible justification $\varphi_k \in \Lambda$ or φ_k following from *Modens Ponens* (see **Definition 12.2.2** for more detail).

It then follows by definition that $\{\varphi_1, \ldots, \varphi_n\} \vdash \varphi(x)$, hence by **Theorem 13.2.1** (i.e. *The Completeness-Lite Theorem*) that $\text{Sen}(\Lambda) \cup \left\{\text{Sen}(\varphi_1), \ldots, \text{Sen}(\varphi_n)\right\} \vDash \text{Sen}(\varphi(x))$. By (a consequence of the Compactness Theorem in sentential logic) we have finite $\{\psi_1, \ldots, \psi_m\} \subset \Lambda$ such that

$$\left\{\text{Sen}(\psi_1), \ldots, \text{Sen}(\psi_m), \text{Sen}(\varphi_1), \ldots, \text{Sen}(\varphi_n)\right\} \vDash \text{Sen}(\varphi(x)).$$

We find that

$$\left\{\text{Sen}(\psi_1), \ldots, \text{Sen}(\psi_m)\right\} \vDash \text{Sen}(\varphi_1) \wedge \cdots \wedge \text{Sen}(\varphi_n) \to \text{Sen}(\varphi(x))$$

since for every valuation mapping $v : \{A_n\}_{n \in \mathbb{N}} \to \{T, F\}$ such that $v\left(\text{Sen}(\psi_j)\right) = T$, for all $1 \leq j \leq m$, since in either case we have

$$v\left(\text{Sen}(\varphi_1) \wedge \cdots \wedge \text{Sen}(\varphi_n)\right) = F \Longrightarrow v\left(\text{Sen}(\varphi_1) \wedge \cdots \wedge \text{Sen}(\varphi_n) \to \text{Sen}(\varphi(x))\right) = T,$$

and

$$\begin{aligned} v\left(\text{Sen}(\varphi_1) \wedge \cdots \wedge \text{Sen}(\varphi_n)\right) = T \Longrightarrow & \ v\left(\text{Sen}(\psi_j)\right) = T, \text{ for all } 1 \leq j \leq m, \\ & \ v\left(\text{Sen}(\varphi_k)\right) = T, \text{ for all } 1 \leq k \leq n, \end{aligned}$$

$$\Longrightarrow v\Big(\text{Sen}(\varphi(x))\Big) = T, \text{ since}$$
$$\Big\{\text{Sen}(\psi_1), \ \dots, \text{Sen}(\psi_m), \text{Sen}(\varphi_1), \ \dots, \text{Sen}(\varphi_n)\Big\} \vDash \text{Sen}(\varphi(x))$$
$$\Longrightarrow v\Big(\text{Sen}(\varphi_1) \wedge \ \cdots \ \wedge \text{Sen}(\varphi_n) \to \text{Sen}(\varphi(x))\Big) = T.$$

It then follows by repeated use of the **Deduction Theorem** that

$$\begin{aligned}&\vDash \text{Sen}(\psi_1) \to \ \cdots \ \to \text{Sen}(\psi_m) \to \text{Sen}(\varphi_1) \wedge \ \cdots \ \wedge \text{Sen}(\varphi_n) \to \text{Sen}(\varphi(x))\\ &\ = \text{Sen}\Big(\psi_1 \to \ \cdots \ \to \psi_m \to \varphi_1 \wedge \ \cdots \ \wedge \varphi_n \to \varphi(x)\Big),\end{aligned}$$

hence we have $\forall x\Big(\psi_1 \to \ \cdots \ \to \psi_m \to \varphi_1 \wedge \ \cdots \ \wedge \varphi_n \to \varphi(x)\Big) \in \Lambda.$

We shall next prove the following claim:

Claim. For all $1 \leq k \leq m$, we have

$$\Gamma \vdash \forall x\Big(\psi_k \to \ \cdots \ \to \psi_m \to \varphi_1 \wedge \ \cdots \ \wedge \varphi_n \to \varphi(x)\Big). \ (13.1)$$

Proof of Claim. By induction on $1 \leq k \leq m$, For the base case $k = 1$, we find (13.1) holds since $\forall x\Big(\psi_1 \to \ \cdots \ \to \psi_m \to \varphi_1 \wedge \ \cdots \ \wedge \varphi_n \to \varphi(x)\Big) \in \Lambda$. If $1 \leq k < m$, then we find (13.1) holds by inductive hypothesis. Moreover, we find **Definition 12.2.1 part 2** *(iii)* that

$$\begin{aligned}&\forall x\Big(\psi_k \to \ \cdots \ \to \psi_m \to \varphi_1 \wedge \ \cdots \ \wedge \varphi_n \to \varphi(x)\Big)\\ &\qquad \to \Big(\forall x\psi_k \to \forall x\Big(\psi_{k+1} \to \ \cdots \ \to \psi_m \to \varphi_1 \wedge \ \cdots \ \wedge \varphi_n \to \varphi(x)\Big)\Big) \in \Lambda.\end{aligned}$$

Then we have

$$\begin{aligned}&\Gamma \vdash \forall x\Big(\psi_k \to \ \cdots \ \to \psi_m \to \varphi_1 \wedge \ \cdots \ \wedge \varphi_n \to \varphi(x)\Big)\\ &\qquad \to \Big(\forall x\psi_k \to \forall x\Big(\psi_{k+1} \to \ \cdots \ \to \psi_m \to \varphi_1 \wedge \ \cdots \ \wedge \varphi_n \to \varphi(x)\Big)\Big),\end{aligned}$$

and in conjunction with (13.1) it follows by *Modens Ponens* that

$$\Gamma \vdash \forall x\psi_k \to \forall x\Big(\psi_{k+1} \to \ \cdots \ \to \psi_m \to \varphi_1 \wedge \ \cdots \ \wedge \varphi_n \to \varphi(x)\Big).$$

Then since $\psi_k \in \Lambda$, we find $\forall x\psi_k \in \Lambda$, so $\Gamma \vdash \ \forall x\psi_k$, and it follows by *Mondens Ponens* that

$$\Gamma \vdash \forall x\big(\psi_{k+1} \to \ \cdots \ \to \psi_m \to \varphi_1 \wedge \ \cdots \ \wedge \varphi_n \to \varphi(x)\big),$$

and the inductive step is complete. □

Using the above *Claim*, we have

$$\Gamma \vdash \forall x\big(\psi_m \to \varphi_1 \wedge \ \cdots \ \wedge \varphi_n \to \varphi(x)\big), \quad (13.2)$$

and using **Definition 12.2.1 part 2** *(iii)*, we have

$$\forall x\big(\psi_m \to \varphi_1 \wedge \ \cdots \ \wedge \varphi_n \to \varphi(x)\big) \to \Big(\forall x\psi_m \to \forall x\big(\varphi_1 \wedge \ \cdots \ \wedge \varphi_n \to \varphi(x)\big)\Big) \in \Lambda.$$

It then follows that

$$\Gamma \vdash \forall x\big(\psi_m \to \varphi_1 \wedge \ \cdots \ \wedge \varphi_n \to \varphi(x)\big) \to \Big(\forall x\psi_m \to \forall x\big(\varphi_1 \wedge \ \cdots \ \wedge \varphi_n \to \varphi(x)\big)\Big),$$

hence by *Modens Ponens* in conjunction with (13.2), we have

$$\Gamma \vdash \forall x\psi_m \to \forall x\big(\varphi_1 \wedge \ \cdots \ \wedge \varphi_n \to \varphi(x)\big). \quad (13.3)$$

Since $\psi_m \in \Lambda$, we find $\forall x\psi_m \in \Lambda$, so $\Gamma \vdash \forall x\psi_m$, and it follows by *Mondens Ponens* in conjunction with (13.3) that we have

$$\Gamma \vdash \forall x\big(\varphi_1 \wedge \ \cdots \ \wedge \varphi_n \to \varphi(x)\big). \quad (13.4)$$

Using **Definition 12.2.1 part 2** *(iii)*, we have

$$\forall x\big(\varphi_1 \wedge \ \cdots \ \wedge \varphi_n \to \varphi(x)\big) \to \big(\forall x(\varphi_1 \wedge \ \cdots \ \wedge \varphi_n) \to \forall x\varphi(x)\big) \in \Lambda,$$

hence

$$\Gamma \vdash \forall x\big(\varphi_1 \wedge \ \cdots \ \wedge \varphi_n \to \varphi(x)\big) \to \big(\forall x(\varphi_1 \wedge \ \cdots \ \wedge \varphi_n) \to \forall x\varphi(x)\big),$$

and it follows by *Modens Ponens* in conjunction with (13.4) that we have

$\Gamma \vdash \forall x(\varphi_1 \wedge \ \cdots \ \wedge \varphi_n) \to \forall x \varphi(x). \quad (13.5)$

Since x is not free in $\varphi_1 \wedge \ \cdots \ \wedge \varphi_n$, we have by **Definition 12.2.1 part 2** *(iv)* that $\varphi_1 \wedge \ \cdots \ \wedge \varphi_n \to \forall x(\varphi_1 \wedge \ \cdots \ \wedge \varphi_n) \in \Lambda$, hence

$\Gamma \vdash \varphi_1 \wedge \ \cdots \ \wedge \varphi_n \to \forall x(\varphi_1 \wedge \ \cdots \ \wedge \varphi_n). \quad (13.6)$

Then by repeated use of $(\wedge I)$, which we proved in **part 2** of this Corollary that since $\Gamma \vdash \varphi_1, \Gamma \vdash \varphi_2, \ \ldots, \Gamma \vdash \varphi_n$, we have $\Gamma \vdash \varphi_1 \wedge \ \cdots \ \wedge \varphi_n$, hence by *Modens Ponens* in conjunction with (13.6), we have $\Gamma \vdash \forall x(\varphi_1 \wedge \ \cdots \ \wedge \varphi_n)$. Then by *Modens Ponens* in conjunction with (13.5), our conclusion of $\Gamma \vdash \forall x \varphi(x)$ is reached.

6. Left as an exercise.

7. Suppose $\Gamma \vdash \varphi^x_t$ and t is substitutable for x in φ. By **Definition 12.2.1 part 2** *(i)*, *(ii)*, we have $\forall x(\neg\varphi) \to \neg\varphi^x_t, \varphi^x_t \to \neg\neg\varphi^x_t \in \Lambda$, hence by **Theorem 13.1.2** and **Corollary 13.1.3**, we have

$$\dfrac{\dfrac{\Gamma \vdash \varphi^x_t \quad \Gamma \vdash \varphi^x_t \to \neg\neg\varphi^x_t}{\Gamma \vdash \neg\neg\varphi^x_t} \to E \qquad \dfrac{\dfrac{\dfrac{\Gamma \vdash \forall x(\neg\varphi) \to \neg\varphi^x_t}{\Gamma; \forall x(\neg\varphi) \vdash \neg\varphi^x_t}\text{Deduction Theorem}}{\Gamma; \neg\neg\varphi^x_t \vdash \exists x \varphi}\text{Contraposition}}{\Gamma \vdash \neg\neg\varphi^x_t \to \exists x\varphi} \to I}{\Gamma \vdash \exists x \varphi.} \to E$$

8. Left as an exercise (Hint: Use **part 6** to show that If x is a variable that doesn't occur free in the hypotheses utilized in some deduction from Γ to $\varphi(x)$, then $\Gamma; \varphi(x) \vdash \forall x \varphi(x)$, then use a contrapositive strategy similar to the previous **part 7** to derive our desired conclusion.) $\square$

We can also formalize the proof system by instead of including an exhaustive list of logical axioms, simply capturing all the logical axioms by instead including the following inference rules:
%CREATE A DEFINITION ON NATURAL DEDUCTION THAT ACCOMPLISHES THIS

$$\frac{}{\Gamma \vdash \alpha}\, Id, \text{ for all } \alpha \in \Gamma$$

$$\frac{\Gamma \vdash \alpha}{\Gamma \cup \Gamma' \vdash \alpha}\, W,$$

$$\frac{\Gamma; \alpha \vdash \beta}{\Gamma \vdash \alpha \rightarrow \beta} \rightarrow I,$$

$$\frac{\Gamma \vdash \alpha \quad \Gamma \vdash \alpha \rightarrow \beta}{\Gamma \vdash \beta} \rightarrow E,$$

$$\frac{\Gamma; \varphi \vdash \beta \quad \Gamma; \varphi \vdash \neg\beta}{\Gamma \vdash \neg\varphi}\, \neg I,$$

$$\frac{\Gamma; \neg\varphi \vdash \beta \quad \Gamma; \neg\varphi \vdash \neg\beta}{\Gamma \vdash \varphi}\, \neg E,$$

$$\frac{\Gamma \vdash \varphi(x)}{\Gamma \vdash \forall x \varphi}\, \forall I,$$ where x is a variable that does not occur in φ or in the preceding sub-deduction of $\Gamma \vdash \varphi(x)$.

$$\frac{\Gamma \vdash \forall x \varphi}{\Gamma \vdash \varphi_x^t}\, \forall E,$$ where t is substitutable for x in φ.

in addition to the equality axioms, which can be written as the inference rules:

$$\frac{}{\Gamma \vdash x = x}\, RI_1,$$

$$\frac{\Gamma \vdash x = y}{\Gamma \vdash y = x}\, RI_2,$$

$\Gamma \vdash x = y \quad \Gamma \vdash y = z$
$----------------RI_3,$
$\Gamma \vdash x = z$

$\Gamma \vdash x_1 = y_1 \cdots \Gamma \vdash \ x_1 = y_n$
$--------------------S_1,$ where t is an n-ary term
$\Gamma \vdash t(x_1, \ \dots, x_n) = t(y_1, \ \dots, y_n)$

$\Gamma \vdash x_1 = y_1 \cdots \ \Gamma \vdash x_1 = y_n \quad \Gamma \vdash \varphi(x_1, \ \dots, x_n)$
$-------------------------------------S_1,$
$\Gamma \vdash \varphi(y_1, \ \dots, y_n)$

where $y_1, \ \dots, y_n$ are free for $x_1, \ \dots, x_n$ in in φ.

%DEFINE PROOF TREES AND DISTINGUISH BETWEEN THE RULES OF INFERENCE FOR PROPOSITIONAL LOGIC AND FIRST ORDER LOGIC

and proofs are in the form of proof trees that are for example formulated as follows:

To show ($\vee I$), where $\alpha \vee \beta$ is defined as $\neg\alpha \rightarrow \beta$, we write the proofs as follows:

$\Gamma \vdash \beta$
$----------W$
$\Gamma; \neg\alpha \vdash \beta$
$-------- \rightarrow I$
$\Gamma \vdash \alpha \vee \beta$

$\Gamma \vdash \alpha$
$----------W, \ ---------------Id,$
$\Gamma; \neg\alpha, \neg\beta \vdash \alpha \qquad \Gamma; \neg\alpha, \neg\beta \vdash \neg\alpha$
$-------------------------\neg E$
$\Gamma; \neg\alpha \vdash \beta$
$-------------- \rightarrow I$
$\Gamma \vdash \alpha \vee \beta$

Theorem 13.1.6. $\Gamma \vdash \varphi$ iff there exists a natural deduction proof tree that proves φ from Γ as described above.

Proof. will write out later in much more detail in a future draft. Basic idea is to cite **Corollary**

13.1.5, as well as the equality axioms for the $\implies$ direction, and for the $\impliedby$ direction, use natural deduction to verify $\vdash \alpha$, for all logical axioms $\alpha \in \Lambda$. $\square$

%State and prove Lemma 24C and Deduction Theorem on page 118

%State and prove equivalence with proof system defined in enderton with other proof system

13.2 Tautological Implication Connection and Soundness Theorem

In some logical texts, we do the completeness theorem in sentential logic, and talk about the proof system in sentential logic (using natural deduction, which I'll talk about in a future draft) uses the inference rules that apply to the sentential part of the system (i.e., scrapping the quantifier and equality rules, or equivalently, the quantifier or equality related logical axioms).

In this approach, we instead show that the proof system agrees with tautological implication, which more or less is soundness/completeness in the sentential setting.

Theorem 13.2.1. *(Completeness-Lite Theorem; Theorem 24B of Enderton)* Let Γ and φ be a set of first order wff's and a wff respectively. Define $\text{Sen}(\Delta) = \left\{\text{Sen}(\alpha) : \alpha \in \Delta\right\}$. Then

$\text{Sen}(\Gamma \cup \Lambda)$ tautologically implies $\text{Sen}(\varphi)$ (i.e. $\text{Sen}(\Gamma \cup \Lambda) \vDash \text{Sen}(\varphi)$ in sentential logic) $\iff \Gamma \vdash \varphi$.

Proof.
%REFER TO ENDERTON

Theorem 13.2.2. *(Soundness Theorem)* $\Gamma \vdash \varphi \implies \Gamma \vDash \varphi$.
%SHOW BETTER PROOF FOR THIS THEOREM

Proof 1. The first proof is given in the book and powerpoint by showing that $\Gamma \vDash \varphi$ if $\varphi \in \Lambda$ and closure of logical implication $\vDash$ under Modens Ponens, which follows immediately by the definition of $\vDash$ applied to $\vDash_{\mathfrak{A}} \psi[s]$, $\vDash_{\mathfrak{A}} \psi \to \varphi[s]$ and for any structure $\mathfrak{A}$ and variable assignment mapping s. $\square$

Proof 2. We can use Theorem 13.1.6 and structural induction on the length of the proof tree that establishes $\Gamma \vdash \varphi$. In the base case where the proof tree is length 1, we either have Id or RI_1. Either way, we have $\varphi \in \Gamma \implies \Gamma \vDash \varphi$ and $\Gamma \vDash \varphi$ if φ is $x = x$.

In the inductive step, we have all the other inference rules, which we can check $\vDash$ follows

suit with closure under those inference rules (which I'll explain in more detail in an update).
%ADD MORE DETAILS TO THIS PROOF

Corollary 13.2.3. The following are equivalent.

1. Sen(Γ) does not tautologically imply $q \wedge \neg q$ for some wff q in the sentential language (as opposed to a first order language where Γ is)

2. There exists a truth assignement function $v : \{A_n\}_{n\in\mathbb{N}} \to \{T, F\}$ satisfying Sen(Γ).

Moreover, if Γ is consistent, then Sen(Γ) is satisfiable in the sense of **part 2** of this corollary (but the converse fails, as shown in the below example).

Example 13.2.4. Note that Sen($\{\forall x(\neg x = x)\}$) $= \{A\}$, which is satisfiable in sentential logic, however, $\forall x(\neg x = x)$ does not have a structure or variable assignment function where it is satisfied. This is where the converse of the second part of **Corollary 13.2.3** is false.

Corollary 13.2.5. Γ is satisfiable $\implies$ Γ is consistent.

%ESTABLISH MORE DETAIL WHEN PROVING THIS PART
Proof. Note that since Γ is satisfiable iff $\Gamma \not\models \beta \wedge \neg\beta$, for some wff β, we have

Γ is satisfiable $\implies \Gamma \not\models \beta \wedge \neg\beta$, for some wff $\beta \implies$ Γ is consistent

by the **Soundness Theorem**. □

Soundness gives us a way of showing when something is consistent, which the proof system simply looking at the definition is otherwise unequipped to do, since to show conistency directly would be be showing that every statement φ proved by Γ is not $\beta \wedge \neg\beta$, doing so is a tedious induction process that we already do through the **Soundness Theorem**, so why not just use the Soundness Theorem.

%GIVE THEOREM 24B

%GIVE SOUNDNESS THEOREM

13.3 Questions

Stephen Herbert asked about whether or not there's a straightforward proof directly by the definition that a theory is consistent?

We have shown that both Sen(Γ) being satisfiable and Γ satisfiable are necessary conditions, so to show failure of consistency, it suffices to show the failure of any of the conditions given in **Corollary 13.2.4** and **Corollary 13.2.6**. And that provides a tool to show consistency.

Soundness gives us a way of showing when something is consistent, which the proof system simply looking at the definition is otherwise unequipped to do, since to show conistency directly would be be showing that every statement φ proved by Γ is not $\beta \wedge \neg\beta$, doing so is a tedious induction process that we already do through the **Soundness Theorem**, so why not just use the soundness Theorem.

%ALSO TALK ABOUT HOW THE PROOF SYSTEM GIVES US AN EASIER WAY TO SHOW UNSATISFIABILITY OF A THEOREM